LAB ASSIGNMENT 01
OBJECT ORIENTED PROGRAMMING

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Course : Object oriented programming with C++
Group : 1MCA3

1. **Write a program (WAP) to display "Hello World" on console display. WAP to implement the following control characters: '\n' is for new line, or you can use endl – cout<<endl<<"message"; '\t' is for tab ; '\a' is an alarm sound; '\r' is carriage return to go to the beginning of the current line**

```
#include <iostream>
using namespace std;
intmain() {
    cout << "Hello world";
    cout << "\nHello world" <<endl;
    cout << "Hello\tworld" <<endl;
    cout << "\aHello" <<endl;
    cout << "Helloo\rworld";
    return 0;
}
```

```
Hello world
Hello world
Hello   world
Hello
worldo
--------------------------------
Process exited after 0.2855 seconds with return value 0
Press any key to continue . . .
```

2. **Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit. [F=9C/5+32]**

```
#include <iostream>
using namespace std;
int main() {
    float C, F;
    cout << "enter temperature :";
    cin >> C;
    F= (9*C)/5+32;
    cout << "temperature in fahrenheit is : " << F;
    return 0;
}
```

```
enter temperature :99
temperature in fahrenheit is : 210.2
--------------------------------
Process exited after 7.394 seconds with return value 0
Press any key to continue . . .
```

3. **WAP to demonstrate for, while, do-while (with all possible variations), like for loop can be demonstrated without giving initialization in for construct or without giving increment in for construct.**

```
#include <iostream>
using namespace std;
int main() {
    int i;
    cout << "loop : ";
    for (i=0; i<10; i++)
    {
        cout << i << " ";
    }
    cout << "\nfor loop without initialization : ";
    i=0;
    for(; i<10; i++);
    {
        cout << i << " ";
    }
    cout << "\nfor loop without increment : ";
    i=0;
    for(; i<10;)
    {
        cout << i<<" ";
        i++;
    }
    cout <<"\nwhile loop : ";
    i=0;
    while(i<10)
    {
        cout << i <<" ";
        i++;
    }
    cout <<"\ndo-while loop : ";
    i=0;
    {
        cout << i<<" ";
        i++;
    }
    while(i<10);
    return 0;
}
```

```
loop : 0 1 2 3 4 5 6 7 8 9
for loop without initialization : 10
for loop without increment : 0 1 2 3 4 5 6 7 8 9
while loop : 0 1 2 3 4 5 6 7 8 9
do-while loop : 0 1 2 3 4 5 6 7 8 9
--------------------------------
Process exited after 0.3446 seconds with return value 0
Press any key to continue . . .
```

4. **Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.**

```
#include <iostream>
using namespace std;
struct Student {
    string name;
    int rollno;
    string degree;
    string hostel;
    float currentCGPA;

    void addDetails()
    {
        cout <<"enter name : ";
        cin>>name;
        cout <<"enter roll no : ";
        cin>>rollno;
        cout <<"enter degree : ";
        cin>>degree;
        cout <<"enter hostel : ";
        cin>>hostel;
        cout <<"enter CGPA : ";
        cin>>currentCGPA;
    }
    void updateDetails()
    {
        cout<<"\nEnter new name : ";
        cin>>name;
        cout<<"\nEnter roll no : ";
        cin>>rollno;
        cout <<"enter degree : ";
        cin>>degree;
    }
    void updateCGPA()
    {
        cout <<"\nenter new CGPA : ";
        cin>>currentCGPA;
    }
    void updateHostel()
    {
        cout <<"\nenter hostel : ";
        cin>>hostel;
    }
    void displaydetails()
    {
        cout<<"\nStudent details"<<endl;
        cout<<"Name : "<<name<<endl;
        cout<<"Roll no. : "<<rollno<<endl;
        cout<<"Degree : "<<degree<<endl;
        cout<<"Hostel : "<<hostel<<endl;
        cout<<"CGPA : "<<currentCGPA<<endl;
    }
};
int main()
{
```

```
    Student s;
    s.addDetails();
    s.displaydetails();
    s.updateCGPA();
    s.updateHostel();
    cout<<"\nAfter update: ";
    s.displaydetails();

    return 0;
}
```

```
Enter name: Mahak Goyal
Enter roll no: 109
Enter degree: MCA
Enter hostel: Q
Enter CGPA: 89

Original Details:
Student Details
Name: Mahak Goyal
Roll No: 109
Degree: MCA
Hostel: Q
CGPA: 89

Enter new name: Simran
Enter new roll no: 108
Enter new degree: BTECh

Enter new CGPA: 85

Enter new hostel: A

After Update:
Student Details
Name: Simran
Roll No: 108
Degree: BTECh
Hostel: A
CGPA: 85
```

5. **Differentiate between private and public access/scope. Perform the question no. 4 with class instead of structure with having the data members private and some member functions in private scope and some in public scope.**

Private	Public
Members can be accessed only inside the class.	Members can be accessed from outside the class.
It provides data hiding.	It allows direct access to members.
If no access specifier is given in a class, members are private by default.	Functions or data declared under public can be accessed using the object.

```
#include <iostream>
using namespace std;

class Student
{
```

```
private:
 string name;
    int rollno;
    string degree;
    string hostel;
    float currentCGPA;

    void updateDetails()
    {
        cout << "\nEnter new name: ";
        cin.ignore();
        getline(cin, name);

        cout << "Enter new roll no: ";
        cin >> rollno;

        cin.ignore();
        cout << "Enter new degree: ";
        getline(cin, degree);
    }

public:
    void addDetails()
    {
        cout << "Enter name: ";
        getline(cin, name);

        cout << "Enter roll no: ";
        cin >> rollno;

        cin.ignore();
        cout << "Enter degree: ";
        getline(cin, degree);

        cout << "Enter hostel: ";
        getline(cin, hostel);

        cout << "Enter CGPA: ";
        cin >> currentCGPA;
    }

    void updateCGPA()
    {
        cout << "\nEnter new CGPA: ";
        cin >> currentCGPA;
    }

    void updateHostel()
    {
        cin.ignore();
        cout << "\nEnter new hostel: ";
        getline(cin, hostel);
    }
```

```
    void displaydetails()
    {
        cout << "\nStudent Details" << endl;
        cout << "Name: " << name << endl;
        cout << "Roll No: " << rollno << endl;
        cout << "Degree: " << degree << endl;
        cout << "Hostel: " << hostel << endl;
        cout << "CGPA: " << currentCGPA << endl;
    }
    void update()
    {
        updateDetails();
    }
};
int main()
{
    Student s;
    s.addDetails();
    cout << "\nOriginal Details:";
    s.displaydetails();
    s.update();
    s.updateCGPA();
    s.updateHostel();
    cout << "\nAfter Update:";
    s.displaydetails();
    return 0;
}
```

```
Enter name: Disha
Enter roll no: 107
Enter degree: MCA
Enter hostel: Q
Enter CGPA: 8.5

Original Details:
Student Details
Name: Disha
Roll No: 107
Degree: MCA
Hostel: Q
CGPA: 8.5

Enter new name: Himanshi
Enter new roll no: 108
Enter new degree: MCA

Enter new CGPA: 7.5

Enter new hostel: Q

After Update:
Student Details
Name: Himanshi
Roll No: 108
Degree: MCA
Hostel: Q
CGPA: 7.5

--------------------------------
Process exited after 252.9 seconds with return value 0
Press any key to continue . . .
```

6. **Create a code snippet that illustrates the following:**
   a. Calling of private member functions inside public member function
   b. Access private member functions inside public member function

(a) Calling of private member functions inside public member function

```
#include <iostream>
using namespace std;
class Student
{
private:
   void privateFunction()
   {
      cout << "Private function is called";
   }
public:
   void publicFunction()
   {
      privateFunction();
   }
};
int main()
{
   Student s;
   s.publicFunction();
   return 0;
}
```

```
Private function is called
--------------------------------
Process exited after 0.2895 seconds with return value 0
Press any key to continue . . .
```

(a) Access private member functions inside public member function

```
#include <iostream>
using namespace std;

class Demo
{
private:
   void privateFunction()
   {
      cout << "This is a private member function";
   }
public:
   void publicFunction()
   {
      cout << "Public function is calling private function:" << endl;
      privateFunction();
   }
};
int main()
{
   Demo obj;
   obj.publicFunction();
   return 0;
}
```

```
Public function is calling private function:
This is a private member function
--------------------------------
Process exited after 0.3223 seconds with return value 0
Press any key to continue . . .
```

7. **Define a class named Complex with properties (real and imaginary) and methods as per following details.**
   void set () to initialize object values.
   void display () to display complex number.
   Complex sum (Complex) or void sum (Complex) to add two complex numbers (objects of Complex class) and return complex_number (object of Complex class) as result.

```
#include <iostream>
using namespace std;

class Complex {
private:
    int real;
 int imaginary;

public:
     void set(int r, int i)
     {
       real = r;
       imaginary = i;


    }
    void display()
    {
       cout << real << " + " << imaginary << "i" << endl;
    }
 Complex sum(Complex c)
    {
       Complex result;

       result.real = real + c.real;
       result.imaginary = imaginary + c.imaginary;
       return result;
    }
};
int main()
{
    Complex c1, c2, c3;
    c1.set(5,3);
    c2.set(2,4);
 c3 = c1.sum(c2);
 cout << "\nFirst complex number: ";
  c1.display();
 cout << "Second complex number: ";
     c2.display();
     cout << "Sum: ";
     c3.display();
     return 0;
    }
```

```
Enter first complex number:
Enter real part: 5
Enter imaginary part: 3

Enter second complex number:
Enter real part: 2
Enter imaginary part: 4

First complex number: 5 + 3i
Second complex number: 2 + 4i
Sum: 7 + 7i

--------------------------------
Process exited after 18.38 seconds with return value
0
Press any key to continue . . .
```

8. **Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.**

```
#include <iostream>
using namespace std;

namespace First {
   int value = 10;
   void display()
   {
      cout << "First namespace" << endl;
      cout << "Value = " << value << endl;
   }
}
namespace Second
{
   int value = 20;
   void display()
   {
      cout << "Second namespace" << endl;
      cout << "Value = " << value << endl;
   }
}
int main()
{
   First::display();
   Second::display();
   return 0;
}
```

```
First namespace
Value = 10
Second namespace
Value = 20

--------------------------------
Process exited after 0.1847 seconds with return value 0
Press any key to continue . . .
```